

Assignment 05 – Part 4: Reflection on Highest Risk Finding and Mitigations

Highest-Risk Finding: Stealthy Fairness Degradation via Label-Flip Poisoning

Across all three parts, the single highest-risk finding is the stealthy fairness degradation caused by the label-flip poisoning attack targeting African American defendants. This attack is particularly dangerous because it causes significant and persistent racial bias (high Adverse Impact Ratio, AIR) while remaining virtually invisible to standard performance monitoring metrics like AUC and traditional drift detection mechanisms like PSI.

Quantified Risk: At a poison rate of just 15%, the AIR for African American defendants jumped from a baseline of 1.961 to 2.134. At a 30% poison rate, the AIR reached 3.010, meaning African American defendants were three times more likely to be falsely classified as high-risk compared to Caucasian defendants among truly low-risk individuals. Crucially, throughout this range (up to 30% poison rate), the model's overall AUC dropped by a negligible ≤ 0.003 percentage points, and feature-based PSI monitoring would not detect any change as only labels were modified. This represents a textbook stealth zone where severe fairness harm is introduced without triggering performance alerts.

Proposed Mitigations

1. Proactive Mitigation: Robust Label Validation and Distribution Monitoring

Action: Implement a proactive system for robust validation and continuous monitoring of label distributions, especially for sensitive groups, during data ingestion and preparation. This involves cross-checking labels against multiple sources, human expert review, and statistical tests to detect unexpected shifts or inconsistencies in label patterns.

Quantified Effect: This mitigation aims to prevent malicious label flips from entering the training data. If successful, it would maintain the Δ AUC at 0.000 (no drop from baseline) and keep the AIR at its baseline of 1.961, preventing the observed degradation to AIR values as high as 3.010. By detecting and correcting even small numbers of flipped labels (e.g., the 23 flipped labels at a 2% poison rate which already led to an AIR of 1.986), it would preserve the model's original fairness characteristics.

Disparate Impact: This mitigation would have a positive disparate impact by directly protecting the targeted African-American group from malicious relabeling that systematically undervalues their risk. By ensuring the integrity of labels, it helps prevent the creation of a biased model, thereby promoting more equitable outcomes for all groups, particularly those at risk of being unfairly disadvantaged.

2. Reactive Mitigation: Real-time Fairness Metric Monitoring (FPR and AIR)

Action: Deploy a real-time monitoring system that continuously tracks key fairness metrics, specifically False Positive Rate (FPR) for each racial group and the Adverse Impact Ratio (AIR) for the deployed model. Set alert thresholds for when AIR deviates outside acceptable bounds (e.g., outside [0.80, 1.25]).

Quantified Effect: This reactive mitigation would detect the attack after it has occurred but before its prolonged impact. The results show that for an African-American-targeted attack, the AIR moves outside the [0.80, 1.25] acceptable range at a 15% poison rate (AIR = 2.134), while AUC remains largely unaffected. A real-time monitor set to alert if AIR exceeds 1.25 would trigger an alarm at this point, allowing operators to investigate and roll back the poisoned model. This prevents the AIR from escalating further to 3.010 as seen at higher poison rates.

Disparate Impact: This mitigation also has a positive disparate impact. It is specifically designed to monitor and protect the fairness of outcomes for different racial groups, particularly the African-American group which was negatively impacted by the poisoning attack. By alerting to unacceptable changes in AIR, it provides a crucial safety net to prevent sustained unfair treatment and facilitates rapid intervention to restore equitable performance.